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High-Resolution Land FWI with Compressive Sensing Acquisition in the Sultanate of Oman

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Summary

Land Full-Waveform Inversion (FWI) holds great potential as confirmed by recent results ([Farooqui et al., 2021](#), and [Guo et al., 2025](#)); however, acquiring high-quality data is crucial for optimal outcomes, particularly in areas with complex near-surface geology. The performance of land FWI can be significantly enhanced using nodal acquisitions that incorporate continuous recordings, longer offsets, and denser sampling, even when decimated or blended. The creation of passive data via interferometry from continuous recordings enables very low-frequency FWI updates, which reduces the risk of cycle skipping at higher frequencies. Furthermore, the use of very long offsets in FWI ensures that the updates are exhaustive, capturing crucial structures and faults, and make high-frequency FWI robust. Using a 2023 3D land survey conducted in the north-eastern part of the Sultanate of Oman, we demonstrate how land FWI can leverage this recent acquisition to deliver superior FWI imaging results compared to those of Pre-Stack Depth Migration (PreSDM), particularly in the most structurally complex area beneath the sand dunes.

Introduction

Despite the significant and promising advancements that have been made in the field of acoustic full-waveform inversion (FWI) in recent years, as documented by [Farooqui et al. \(2021\)](#) and [Guo et al. \(2025\)](#), the application of this technology in land environments of the Middle East continues to pose substantial challenges. The inherent difficulties associated with land FWI, which are typically linked to the presence of complex near-surface geological structures, are often magnified in this region, resulting in a heightened level of scattered noise, multiples, surface waves, and mode conversions. These geological complexities can significantly compromise the accuracy and reliability of FWI, thereby limiting its effectiveness in providing high-resolution images of the subsurface.

However, recent efforts to design and deploy cutting-edge, high-end acquisition systems specifically tailored to land FWI have yielded new opportunities for improving the resolution and accuracy of subsurface imaging. The development of acquisition systems that incorporate broadband, dense, and long offsets as well as continuous recordings, as discussed by [Mahrooqi et al. \(2012\)](#), has been shown to enhance the quality

and resolution of the acquired data. Furthermore, the integration of innovative techniques such as blended and compressive sensing (CS) methodologies, as explored by [Blymyer et al. \(2021\)](#), has contributed to making these advanced acquisition systems more commercially viable and accessible, thereby expanding their potential applications in the field of land FWI.

In this study, we present a case study conducted in the northern region of the Sultanate of Oman, an area characterized by complex geological features, including a thrust system situated beneath a gravel plain and sand dunes. The data acquisition for this study was designed to combine the benefits of blending, compressive sensing (with a 75% CS ratio), and ultra-long offsets, which provides a comprehensive dataset for the application of FWI. We demonstrate how a specifically designed workflow, tailored to leverage the advantages of nodal acquisition with ultra-long offsets and continuous recording, enables the successful application of high-resolution land FWI. A key aspect of this workflow is its ability to recover ultra-low frequency data by interferometry, with frequencies extending below the minimum sweep frequency of 1.5 Hz. Moreover, the deeply penetrating diving-wave energy generated by the long offsets plays a crucial role in facilitating the construction of a robust and accurate model with FWI. Finally, we also demonstrate the potential for high-resolution land FWI to provide detailed and accurate images of complex subsurface structures, even in challenging geological environments such as those encountered in the Middle East.

Data acquisition and geological context

The survey area of interest is situated in the eastern portion of the Ghaba salt basin, which is located in the northern region of the Sultanate of Oman. Geographically, it is positioned on the gravel plain near the Hajar Mountains with an eastern extension into the Wahiba sand dunes. This region is characterized by a complex geological setting, marked by the presence of a thrust system, as documented by [Rollinson et al. \(2014\)](#). The thrust system is a dominant structural feature that divides the area into two distinct geological domains. To the north of the main thrust, the exposure of Jurassic Hawasina nappes provides a unique window into the region's geological past, offering valuable insights into the tectonic evolution of the area. In contrast, south of the thrust, modern sediments are being deposited, reflecting the ongoing geological processes that are shaping the region.

The area is also characterized by the presence of northwest-southeast (NW-SE) oriented normal faults, which are located to the south of the Semail thrust front. These faults are a notable feature of regional geology, and their orientation is also reflected in the alignment of certain surface features, such as wadis. The presence of these faults and the thrust system contributes to the complexity of regional geology, making it a challenging area for seismic data acquisition and interpretation.

In 2023, a new 3D seismic campaign was conducted, covering an area of approximately 2100 square kilometers. The survey design was tailored to the specific terrain conditions with a 75% compressive sensing design in the gravel plain area on the western side of the survey and a cross-spread design over the sand dunes on the eastern side ([Figure 1](#)). The sand dunes in the area have an average height of 60 meters, which presented a significant challenge for the velocity model building. To address this challenge, time static corrections were applied to compensate for the travel time propagation in the sand, ensuring consistent topography based on the underlying gravel plain for the entire project area. These corrections were computed from a regional absorption coefficient dependent on the sand thickness, as described by [Hanafy et al. \(2020\)](#).

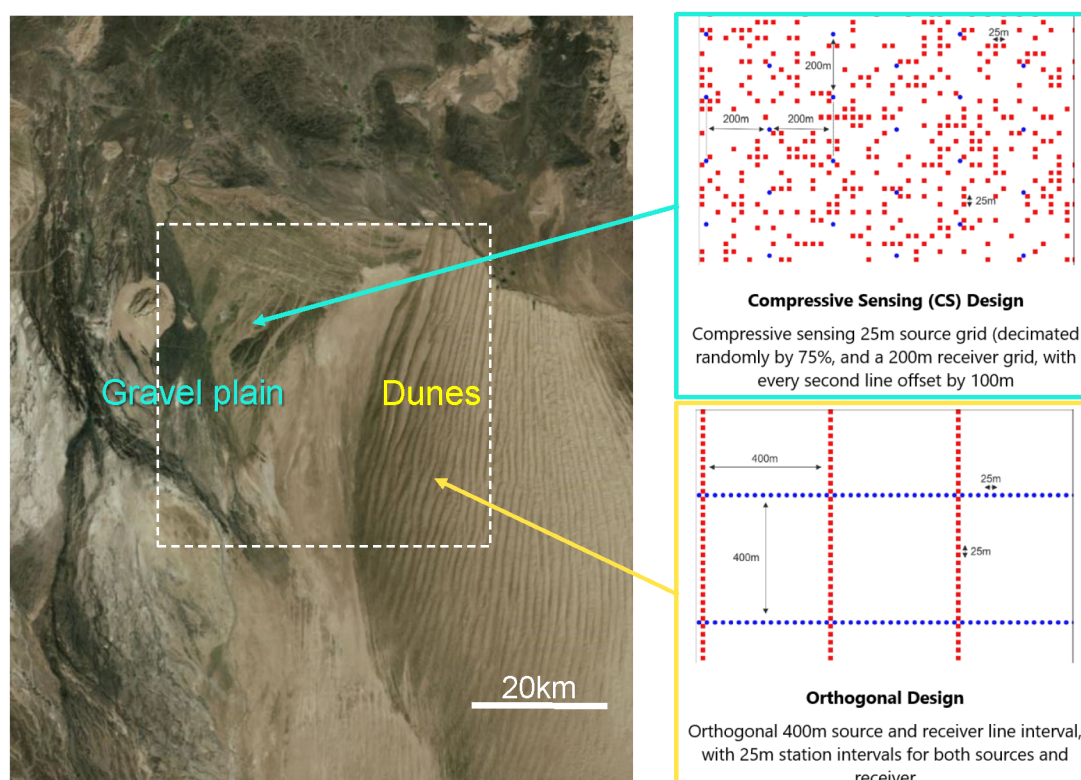


Figure 1—Survey polygon (in white) over a satellite image with acquisition designs shown according to the terrain: carpet shooting with compressive sensing over the gravel plain (west) and cross-spread design over the sand dunes (east).

The seismic acquisition was nodal in nature, with a maximum offset of 40 kilometers and a consistent offset of 15 kilometers available throughout the survey area. The CS design consisted of a 50-meter interval between shot lines and source stations, and 200-meter receiver line and station spacing. In contrast, the cross-spread design had a 400-meter interval between lines and 25-meter interval between stations for both shots and receivers. The active source used for the survey was vibroseis, with a sweep frequency range of 1.5 Hz to 86 Hz. The acquisition parameters were carefully designed to optimize the quality of the seismic data, considering the complex geological setting and the challenging terrain conditions. The resulting dataset provides a high-quality foundation for subsequent seismic processing and FWI.

Impact of ultra-low frequencies and long offsets

The accuracy of the initial velocity model is a crucial factor for the success of FWI, as it provides the foundation for subsequent model updates and refinement. However, the dependency on the initial model can be significantly relaxed by starting the inversion process from the lowest available frequencies. This approach reduces the risk of cycle skipping and improves the overall convergence of the inversion process.

The ambient noise recorded continuously during this acquisition offered a unique opportunity to use interferometry to build virtual shots with frequencies below those generated by the active sources. As demonstrated by [Le Meur et al. \(2020\)](#), continuous recordings can register passive sources such as the sea waves hitting the shore ([Figure 2a](#)), which were found to be a prominent natural source in this study. The continuous recordings successfully captured this energy, providing a valuable dataset for the reconstruction by interferometry of virtual shot recordings.

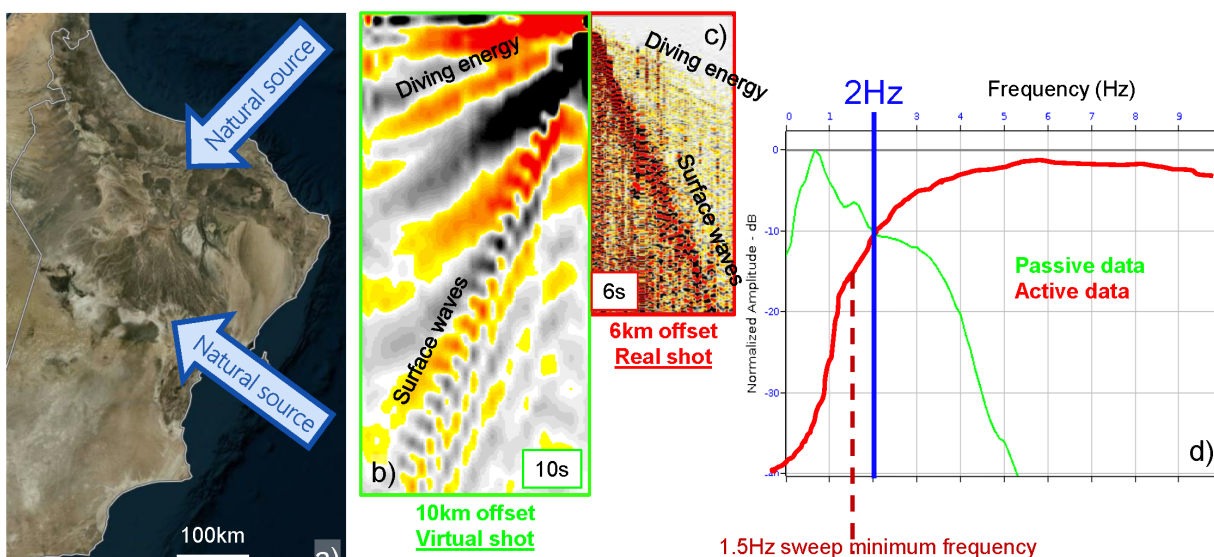


Figure 2—Satellite image displaying the direction of arrival of the sea waves hitting the shore and registered as ambient noise by the continuous recordings (a), one virtual shot built by interferometry (b) compared to the real shot (c) at the same location, both presenting complementary frequency spectra (d). Passive data are characterized by lower frequency content (starting at 0.5Hz) than the minimum sweep frequency (1.5Hz).

The data reconstructed by interferometry from the continuous recordings exhibited several notable characteristics. Firstly, they were characterized by longer offsets than the nominal active data (Figures 2b and 2c), which is a significant advantage for FWI. Secondly, the passive data had a lower frequency content than the sweep minimum frequency of 1.5 Hz (Figure 2d), which is a critical aspect for the initial phases of the FWI model updates. The ability to access frequencies below the minimum sweep frequency of the active sources enables the inversion to start from a more robust and stable foundation, which reduces the risk of cycle skipping and improving the overall convergence of the process.

Furthermore, the interferometry-based data presented not only strong ground roll but also body wave energy (Figure 2b), which is a valuable component of the seismic wavefield. The whole wavefield of the passive data, including both ground roll and body waves, was inverted for the initial phases of the FWI model updates, enabling the construction of a more accurate and robust initial velocity model for higher frequency FWI updates.

The initial velocity model was constructed through a multi-step process, involving the integration of near-surface and deeper velocity information. Firstly, the multi-wave inversion (MWI) approach, as described by Bardainne (2018), was employed to generate a nearsurface velocity model. This involved the combination of first-break information, picked up to a maximum offset of 5 km, with surface-wave dispersion curves. The fundamental mode of the dispersion curves was carefully picked over a frequency range of 0.5 to 8 Hz, utilizing both passive and active data. The resulting MWI near-surface velocity model allowed updating the first few hundreds of meters below the topography, providing a robust near-surface model for subsequent velocity model updates.

To further improve the initial velocity model, both diving waves and surface waves from the passive data were inverted using acoustic FWI for V_p update of diving waves and elastic FWI, as described by Leblanc et al. (2022) and Sedova et al. (2024), for V_s update of surface waves over the frequency range of 0.5 to 2 Hz. A regional V_p/V_s ratio, starting at 2.5 below the topography and gradually decreasing down to 2, was used to convert V_p and V_s models. This very low frequency update made it possible to capture the first fast layer between the Natih and Gharif formations, as illustrated in Figures 3a and 3b.

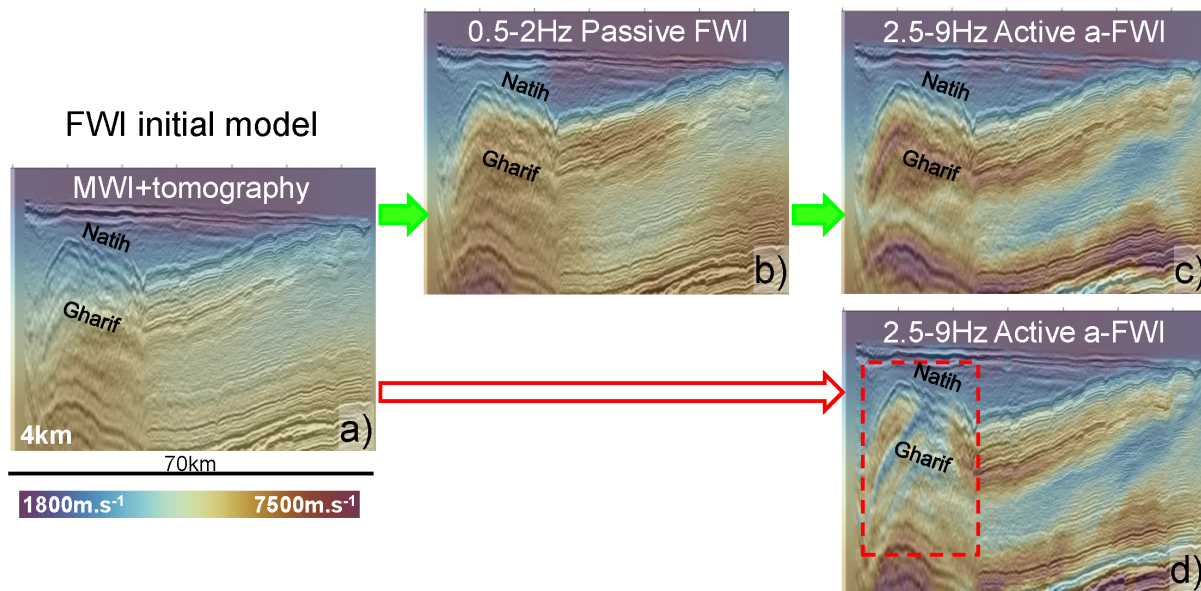


Figure 3—Starting from the same initial model (in a), the inclusion of the 0.5-2 Hz FWI update with passive data (in b) provides a more conformal FWI result when using active data from 2.5 to 9 Hz (in c). In d, the result without the passive data FWI update is affected by instabilities from cycle skipping (red rectangle).

The benefits of this very low frequency update are evident when used as a starting model for subsequent acoustic FWI from 2.5 to 9 Hz using active data, as shown in Figure 3c. In contrast, starting the 2.5-9 Hz FWI update from the initial model without the very-low frequency FWI update resulted in cycle-skipping in some areas, leading to incorrect velocity updates, such as within the red rectangle of Figure 3d. The incorporation of the very-low frequency FWI update significantly reduced the risk of cycle skipping at higher frequencies, highlighting the importance of this step in the velocity model building process.

Prior to acoustic FWI, the active data after deblending underwent a series of pre-conditioning steps to enhance the signal-to-noise ratio of the low frequencies and remove undesired noise, primarily surface waves. The pre-conditioning workflow consisted of several key steps optimized for both acquisition design and the sand dunes. The FWI data preparation included the application of surface-consistent amplitude and phase correction operators to stabilize the data wavelet, FK noise attenuation for velocities less than 2000 m/s to remove strong surface wave energy, and 4D low rank random noise attenuation applied in cross spread domain, as described by Sternfels et al. (2015), to further denoise and enhance the continuity of diving and reflected energies. These pre-conditioning steps were essential in preparing the active data for acoustic FWI, ensuring that the subsequent velocity model updates were accurate and reliable. As both acquisition datasets are not regularized together before FWI, their different offset distributions and their different signal to noise ratios and acquisition footprints necessitated some additional adjustments on the single updated velocity model at some frequency steps.

The nodal acquisition employed in this study provided a unique opportunity to utilize seismic data with maximum offsets of up to 40 km in certain areas. This extensive offset range allowed us to investigate the impact of the acquired long offsets on the FWI update. The complex geology of the region, as illustrated in Figure 4b, posed a significant challenge for seismic imaging. However, by utilizing the full available offsets, we were able to better resolve the subsurface structure and stabilize the 9 Hz acoustic FWI velocity update.

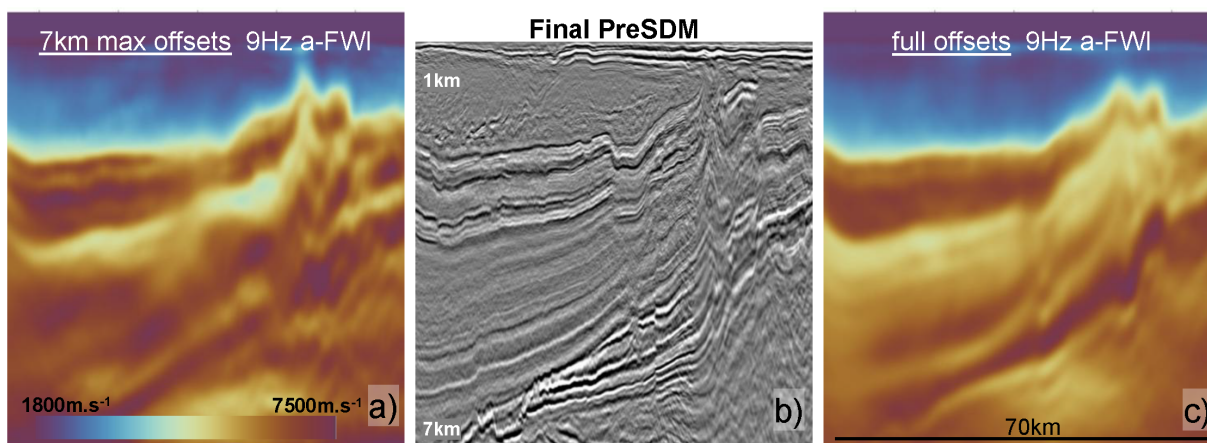


Figure 4—2.5-9 Hz acoustic FWI results using maximum offset of 7 km (a) and using all available offsets (c), which generates a clear layered update, conformal to the layers in the PSDM final image (b).

The incorporation of long offsets in the FWI process yielded more continuous layers, main faults, and structures that could be imaged down to a depth of 7 km, as shown in Figure 4c. This improvement in imaging depth and resolution can be attributed to the increased penetration of diving wave energy enabled by the long-offset data. This provided more comprehensive subsurface sampling and played a crucial role in stabilizing the acoustic FWI, delivering a robust velocity model for subsequent higher frequency FWI updates using reflected energy. In contrast, using data with offsets smaller than 7 km in the FWI leads to shallower update, and this update has reduced reliability, as illustrated in Figure 4a. Using data with a maximum offset of 7 km only allowed to retrieve the first strong velocity at the Natih level, with the deeper part of the model only updated by reflected energy.

High-resolution FWI

Recent studies, such as that of Guo et al. (2025), have demonstrated the significant benefits of high-frequency land Full-Waveform Inversion (FWI) and FWI Imaging in improving the imaging of complex geological features. Given the structurally complex geology of the area under investigation, our primary objective was to obtain a high-resolution acoustic FWI model that could be used for migration and FWI Imaging, thereby facilitating the evaluation of prospects in this exploration survey.

To achieve this goal, we employed a velocity model building flow that involved multiple passes of acoustic FWI interleaved with

Residual Move-Out (RMO) tomography for velocity and anisotropy updates using wells (Lambare et al., 2007). This iterative process was necessary due to the near-surface and structural diversity of geology over the entire area, as well as the absence of legacy velocity information. The final pass of acoustic FWI was conducted over the frequency range of 3-30 Hz, utilizing full offsets at low frequencies to leverage diving wave energy and gradually limiting offsets to 4 km at the highest frequencies to prioritize reflected energy and enhance resolution.

Kirchhoff PSDM and FWI Images were computed with the 30 Hz FWI velocity model (Figure 5f). Figure 5a shows that the PSDM has a high noise level in the region below the sand dunes (on the right) compared to the region without sand dunes (on the left). In contrast, FWI Imaging (Figure 5b) provides a much clearer representation, highlighting a distinct dome shape, sharper structural features, and major faults while being minimally affected by sand dunes. A comparison of Figures 5c and 5d reveals that FWI Imaging displays more consistent events, including those that are steeply dipping, as well as a clear layering, resulting in improved imaging in the most intricate geological area. This enhancement can be attributed to the fact that the FWI used a wider range of data to update velocity (Figures 5e and 5f) instead of just primary reflections to construct the model. The well matching demonstrated in Figure 5g for the Well-A well (located in Figure

5b) accurately identifies significant deep events in the FWI Image (black wiggles) that give a good match with the synthetic trace (red wiggles).

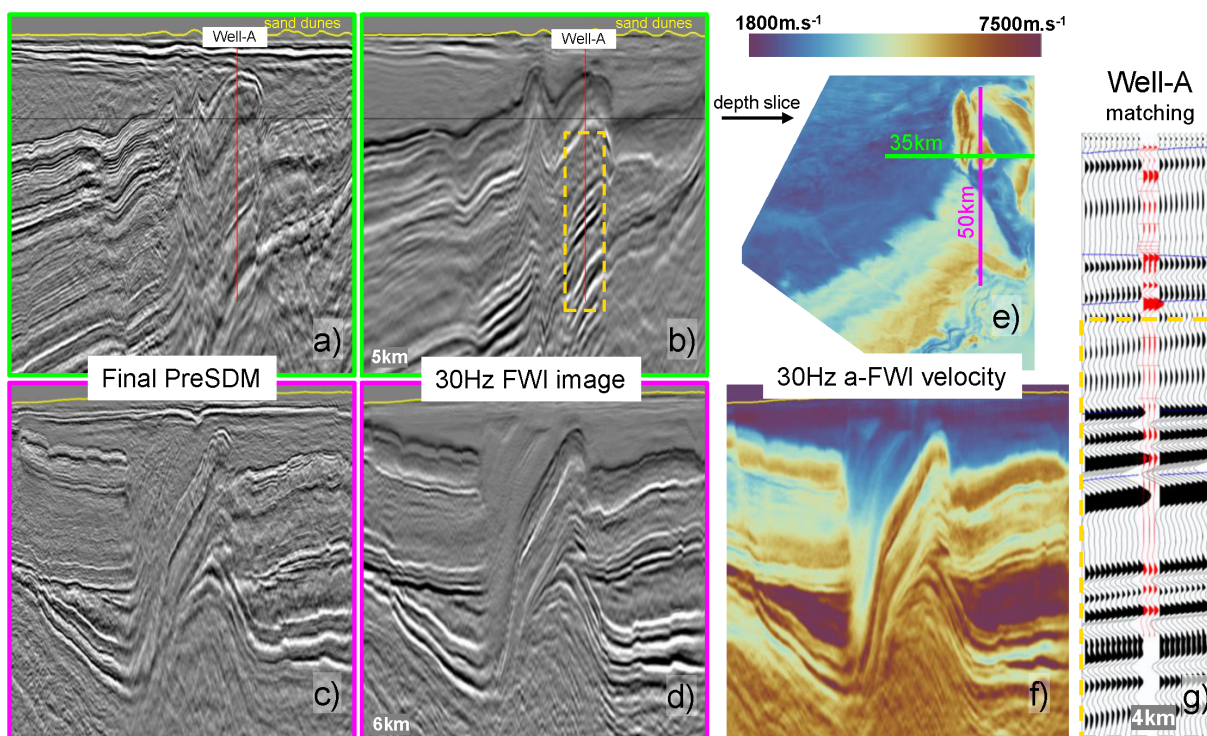


Figure 5—In the most structurally complex area below the sand dunes (yellow topography line in a, b, c and d), the 30 Hz FWI Image (b and d) shows notable continuity of the key events and an interpretable low frequency layering, where the final PSDM image has some residual noise and artefacts (a and c). The 30 Hz FWI Image matches well with the synthetic trace of Well-A (g). The 30 Hz a-FWI velocity shows a clear fine layering in section view (f) and high lateral resolution in depth slice view (e).

Conclusions

The land case study presented here is from a modern acquisition. It demonstrates how an efficient velocity model building workflow based on FWI can be developed to achieve successful high-resolution FWI model and Imaging. It shows the potential offered by a long-offset nodal acquisition with continuous recordings. First, interferometry enables building ultra-low frequency virtual data containing P-wave and surface wave energies, inverted by acoustic and elastic FWI, to help mitigate cycle-skipping at higher frequencies. Second, long offsets allow to delineate a complex layering of intricate structures and primary faults, thanks to deeply penetrating diving wave energy successfully used by FWI. Finally, pushing acoustic land FWI to high resolution with frequencies up to 30 Hz shows that the resulting FWI Imaging surpasses conventional Kirchhoff PSDM, particularly in the complex area beneath the sand dunes.

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